

YEAR 7

MATHS & SCIENCE

What pupils in England are expected to learn as they enter Key Stage 3

Government-source edition • Department for Education / GOV.UK only

The crucial point

The government specifies the curriculum for Key Stage 3 (Years 7–9), not a single national Year 7 teaching order. Schools publish their own year-by-year curriculum and may sequence the statutory content differently.

Prepared as a clear reference for families supporting pupils entering Year 7.

1. How to use this guide

This guide answers two different questions that are often confused:

- What does the Department for Education require pupils to learn during Key Stage 3?
- What should a parent or tutor watch for during the first year of secondary school?

The first is statutory. The second depends on the school's sequencing. For that reason, the curriculum maps below show the full KS3 entitlement, while the "Year 7 lens" identifies the foundations that make later KS3 learning possible rather than claiming a government-mandated Year 7 order.

The government's three big mathematical aims

- Fluency — secure, flexible command of number, algebra and representations.
- Reasoning — making connections, testing conjectures, justifying and proving.
- Problem solving — choosing mathematics for routine and unfamiliar multi-step problems.

The government's three big science aims

- Scientific knowledge and conceptual understanding across biology, chemistry and physics.
- Understanding scientific methods, processes and enquiry.
- Using scientific knowledge to understand applications and implications of science.

Parent takeaway: success in Year 7 is not simply "covering topics". It is the beginning of a three-year progression in knowledge, reasoning, vocabulary, practical competence and independence.

2. Mathematics — the KS3 curriculum map

The statutory KS3 mathematics programme is organised into six content domains, underpinned by working mathematically. The concise descriptions below are faithful summaries of the DfE requirements.

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
NUMBER	Place value with integers and decimals; positive and negative numbers; fractions and mixed numbers; four operations; primes, factors, multiples, HCF/LCM and prime factorisation; powers and roots; standard form; fraction–decimal–percentage equivalence; percentages; units, rounding, estimation and calculator use.
ALGEBRA	Algebraic notation and vocabulary; substitution; simplifying and manipulating expressions; formulae and rearranging; linear equations; coordinates; graphs of linear and quadratic functions; gradients and intercepts; graphical solutions; sequences including arithmetic and geometric patterns and nth-term work.
RATIO & PROPORTION	Standard-unit conversions; scale factors, diagrams and maps; fractions as comparisons; ratio notation and division in a ratio; multiplicative relationships; links between ratio, fractions and linear functions; percentage change and simple interest; direct and inverse proportion; compound units such as speed, unit price and density.
GEOMETRY & MEASURES	Perimeter, area and volume; circles and composite shapes; scale drawings; ruler-and-compass constructions; geometric notation; congruence and similarity; transformations; angle facts and parallel lines; polygon angle sums; Pythagoras and introductory trigonometric ratios; 3-D properties and geometric reasoning.
PROBABILITY	Experimental frequency; fairness and equally/unequally likely outcomes; the 0–1 probability scale; probabilities summing to 1; systematic enumeration with tables, grids and Venn diagrams; sample spaces and theoretical probabilities for single and combined events.
STATISTICS	Representing and comparing distributions; discrete, continuous and grouped data; mean, median, mode, range and outliers; frequency tables and charts; bar and pie charts; interpreting relationships between two variables and representing them with scatter graphs.

Working mathematically — what sits underneath every topic

- Develop fluency: calculate accurately, use algebra confidently, and move between numerical, algebraic, graphical and diagrammatic forms.
- Reason mathematically: identify relationships, use ratio and proportion, test conjectures, begin deductive reasoning and judge what conclusions data can support.
- Solve problems: tackle multi-step and unfamiliar problems, use mathematics in financial contexts, model situations and select appropriate techniques.

3. Mathematics — the Year 7 transition lens

Because the DfE does not prescribe a national Year 7 order, use this as a readiness and support map. These are the foundations a pupil should become increasingly secure with as KS3 begins.

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
NUMBER CONFIDENCE	Integers, place value, negative numbers, calculation fluency, fractions/decimals/percentages, factors and multiples, powers and roots.
ALGEBRA BEGINS TO BECOME A LANGUAGE	Using letters, terms and expressions; substituting values; collecting like terms; forming and solving simple equations; seeing patterns as general rules.
PROPORTIONAL THINKING	Ratio, scale, unit conversions and percentage reasoning — the bridge into science, finance, graphs and later GCSE work.
GEOMETRY WITH REASONS	Angle facts, properties of shapes, coordinates, transformations, perimeter/area/volume and explaining why a result follows.
DATA & CHANCE	Reading and creating graphs and charts, averages and spread, simple probability, and learning not to overclaim from data.
MATHEMATICAL HABITS	Showing working clearly, selecting a method, checking reasonableness, explaining an argument, learning from errors and persevering with unfamiliar problems.

A simple parent check

- Can your child explain a method, not merely produce an answer?
- Are fractions, decimals, percentages and negative numbers becoming dependable rather than fragile?
- Can they translate between words, tables, graphs, diagrams and algebra?
- Can they recognise when a problem is additive, multiplicative or proportional?
- Can they show enough working for someone else to follow their reasoning?

4. Science — the KS3 curriculum map

KS3 science is explicitly disciplinary: biology, chemistry and physics. Across all three, pupils must also work scientifically — planning, measuring, analysing, evaluating and communicating with increasing precision.

Working scientifically

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
SCIENTIFIC ATTITUDES	Objectivity; accuracy and precision; repeatability and reproducibility; how theories change with evidence; publication and peer review; risk evaluation.
EXPERIMENTAL SKILLS	Ask questions; make predictions; identify variables; select and carry out suitable enquiries; use apparatus safely; measure and record; evaluate methods; use sampling.
ANALYSIS & EVALUATION	Apply mathematics; present data in tables and graphs; identify patterns; draw conclusions; relate evidence to predictions; recognise random/systematic error; generate further questions.
MEASUREMENT	Use SI units and chemical nomenclature; use and derive simple equations; perform calculations; carry out basic statistical analysis.

5. Biology — what pupils must encounter across KS3

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
CELLS & ORGANISATION	Cells as the basic unit of life; microscopy; functions of major cell structures; plant versus animal cells; diffusion; unicellular adaptations; organisation from cells to tissues, organs, systems and organisms.
SKELETON & MUSCLES	Functions of the skeleton; biomechanics; forces produced by muscles; antagonistic muscle pairs.
NUTRITION & DIGESTION	Components of a healthy diet and their roles; energy needs; consequences of dietary imbalance; digestive organs and adaptations; enzymes as biological catalysts; gut bacteria; plant nutrition and water/mineral uptake.
GAS EXCHANGE	Human gas-exchange structures and functions; breathing using a pressure model; lung-volume measurements; effects of exercise, asthma and smoking; stomata in plants.
REPRODUCTION & HEALTH	Human reproductive systems, menstrual cycle at an introductory level, gametes, fertilisation, gestation and birth; maternal lifestyle and the foetus; plant reproduction, pollination, fertilisation, seed/fruit formation and dispersal; effects of recreational drugs.
PHOTOSYNTHESIS & RESPIRATION	Reactants/products and word equations; photosynthesis as the basis of food/energy stores and atmospheric balance; leaf adaptations; aerobic and anaerobic respiration, including fermentation, and their biological significance.
ECOLOGY, GENETICS & EVOLUTION	Food webs and interdependence; pollination and food security; environmental effects and toxic accumulation; heredity, chromosomes, genes and DNA; variation; natural selection; extinction; biodiversity and gene banks.

6. Chemistry — what pupils must encounter across KS3

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
PARTICLE MODEL	Solids, liquids and gases explained through particles; gas pressure; changes of state.
ATOMS, ELEMENTS & COMPOUNDS	A simple atomic model; distinguishing atoms, elements and compounds; symbols and formulae; conservation of mass in physical and chemical change.
PURE & IMPURE SUBSTANCES	Pure substances, mixtures and dissolving; diffusion; filtration, evaporation, distillation and chromatography; identifying purity.
CHEMICAL REACTIONS	Reactions as rearrangements of atoms; formulae and equations; combustion, decomposition, oxidation and displacement; acids and alkalis, pH and indicators; neutralisation; acid reactions with metals and alkalis; catalysts.
ENERGETICS	Qualitative energy changes during changes of state and in exothermic/endothermic reactions.
PERIODIC TABLE	Patterns in physical and chemical properties; Mendeleev's organising principles; groups and periods; metals and non-metals; using periodic patterns to predict reactions; metal/non-metal oxides and acidity.
MATERIALS	Reactivity series including carbon; extraction of metals from oxides using carbon; properties of ceramics, polymers and composites.
EARTH & ATMOSPHERE	Composition and structure of Earth; rock cycle; finite resources and recycling; atmosphere composition; human carbon dioxide production and climate impact.

7. Physics — what pupils must encounter across KS3

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
ENERGY	Energy in food and appliances; power ratings; energy transferred; domestic fuels, bills and resources; energy transfers by heating, radiation, motion, circuits, deformation, metabolism and combustion; conservation of energy in systems.
MOTION & FORCES	Speed = distance ÷ time; distance–time graphs; relative motion; contact/non-contact forces; force diagrams; balanced/unbalanced forces; moments; friction and drag; springs and Hooke’s law; work and deformation; gravity, magnetism and static forces; pressure, upthrust and equilibrium.
WAVES & SOUND	Water waves, reflection and superposition; frequency in hertz; echoes, absorption and reflection; sound needing a medium; vibrations; longitudinal sound waves; hearing ranges; waves transferring energy and information.
LIGHT	Light in a vacuum; reflection, scattering and absorption; ray models for mirrors, pinhole cameras, refraction and convex lenses; the eye; light energy effects; colour, frequency, white light and prisms.
ELECTRICITY & MAGNETISM	Current in series and parallel circuits; charge flow; potential difference and resistance; conductors and insulators; static charge and electric fields; magnetic poles and field lines; Earth’s field; electromagnets and basic DC motor principles.
MATTER	Physical versus chemical change; conservation and reversibility; density differences; Brownian motion; diffusion; particle arrangements/motion and changes of state; atoms and molecules; temperature, particle motion and internal energy.
SPACE	Weight and gravitational field strength; Earth–Moon–Sun gravity; Sun as a star, our galaxy and other galaxies; seasons and axial tilt; changing day length; light-year as astronomical distance.

8. Science — the Year 7 transition lens

A strong first year of KS3 science should begin moving the pupil from primary-school description toward secondary-school explanation. Whatever order the school chooses, watch for these shifts:

DOMAIN	WHAT THE DfE EXPECTS ACROSS KEY STAGE 3
FROM SEEING TO MODELLING	Using particles, cells, forces, energy and waves as models that explain observations rather than merely naming facts.
FROM “FAIR TEST” TO VARIABLES	Identifying independent, dependent and control variables; choosing an enquiry type rather than following a recipe.
FROM MEASURING TO JUDGING DATA	Units, tables and graphs; patterns; conclusions; reliability; sources of error; sensible improvements.
FROM EVERYDAY WORDS TO SCIENCE LANGUAGE	Precise vocabulary, symbols, units, formulae, nomenclature and correctly structured explanations.
FROM SUBJECT ISLANDS TO CONNECTIONS	Seeing how maths supports science and how biology, chemistry and physics connect — for example particles, energy transfer, structure/function and quantitative relationships.

Questions parents can ask without turning home into another classroom

- What idea is this experiment testing?
- What did you change, measure and keep the same?
- What does the graph actually show?
- Which scientific model explains what happened?
- What evidence would make you change your explanation?
- Where did you use mathematics in this science topic?

9. The parent support plan — one page to remember

The aim is not to pre-teach all of KS3. It is to keep the foundations strong enough that school learning can compound.

WATCH	SUPPORT	ASK
Maths fluency	Short, regular practice of weak foundations	“Can you explain why that method works?”
Maths reasoning	One unfamiliar problem, not twenty repetitive ones	“How else could you represent this?”
Science knowledge	Recall key vocabulary, models and equations	“What is the big idea underneath this topic?”
Science enquiry	Discuss variables, measurements and graphs	“How trustworthy is the evidence?”
School alignment	Check the school’s published Year 7 curriculum each term	“Which KS3 strand are you doing now?”

A useful rhythm: know what the school is teaching now, identify one fragile prerequisite, strengthen it, then use retrieval and explanation to keep the new learning alive.

10. What to obtain from your child’s school

- The school’s Year 7 maths curriculum map or scheme overview.
- The school’s Year 7 science curriculum map or scheme overview.
- The order of units by term.
- How and when pupils are assessed.
- The school’s calculation, homework and revision expectations.
- Any set-specific or pathway-specific differences.

This matters because the DfE explicitly gives schools flexibility over when content is introduced within a key stage, while requiring schools to publish their curriculum year by year.

11. Official government sources used

No commercial curriculum, tutoring website, exam-board scheme or school scheme of work was used to define the curriculum content in this guide.

Department for Education. National curriculum in England: mathematics programme of study — key stage 3.

Ref: DFE-00179-2013

Published 11 September 2013; GOV.UK page updated 28 September 2021.

<https://www.gov.uk/government/publications/national-curriculum-in-england-mathematics-programmes-of-study>

Department for Education. National curriculum in England: science programme of study — key stage 3.

Ref: DFE-00185-2013

Published 11 September 2013; GOV.UK page updated 6 May 2015.

<https://www.gov.uk/government/publications/national-curriculum-in-england-science-programmes-of-study>

Department for Education. National curriculum: secondary curriculum.

Ref: DFE-00183-2013

Published 11 September 2013. Statutory secondary programmes of study for Key Stages 3 and 4.

<https://www.gov.uk/government/publications/national-curriculum-in-england-secondary-curriculum>

Department for Education. Teaching mathematics at key stage 3.

Non-statutory guidance

Published 28 September 2021. Used only to clarify progression and curriculum planning, not to invent a Year 7 statutory syllabus.

<https://www.gov.uk/government/publications/teaching-mathematics-at-key-stage-3>

Final orientation

For a pupil entering Year 7, the most useful mental model is this: Year 7 is the opening movement of Key Stage 3, not a sealed syllabus of its own. The destination is the full KS3 entitlement above; the school decides the route. Our job as parents and educators is to know the destination, inspect the school's route, and make sure the pupil's foundations are strong enough for the journey.